Formal Cover Letter for Journal Submission

To:
The Editor-in-Chief
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From:
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Subject: Submission of Original Research Article: *"InSituGame: In Situ Spatial Evolutionary Game Assay for Inferring Tumor Subclonal Payoff Matrices and Exploring Adaptive Therapy in Intact Human Biopsies"*

Dear Editor-in-Chief and Editorial Board Members,

I am pleased to submit our original research manuscript entitled **"InSituGame: In Situ Spatial Evolutionary Game Assay for Inferring Tumor Subclonal Payoff Matrices and Exploring Adaptive Therapy in Intact Human Biopsies"** for consideration as an Original Article in *Bioinformatics*.

Background and Scientific Rationale

For over two decades since the founding formulations of evolutionary oncology, **evolutionary game theory (EGT)** has provided a compelling mathematical foundation to counter acquired cancer drug resistance. By modulating therapy to maintain a competing subpopulation of treatment-sensitive cells, adaptive therapy prevents competitive release of refractory subclones, substantially prolonging progression-free survival in clinical pilot trials.

However, translation to bedside oncology has been constrained by a fundamental bottleneck: **how can we estimate the subclonal interaction payoff matrix (A) directly from a patient's intact clinical biopsy?** Historically, researchers were limited to either postulating mathematical heuristics or measuring interactions in artificial 2D in vitro co-cultures stripped of the native 3D extracellular matrix, stroma, and physical microenvironmental gradients.

In this work, we address this translational bottleneck by developing **InSituGame** (*In Situ Spatial Evolutionary Game Assay*), an exploratory computational framework with five core contributions: 1. **Mathematical Foundation & Ergodicity:** We formulate the **Space-for-Time Ergodic Mapping Framework (Proposition 1)**, establishing how single-cell-cycle transcriptional snapshots across spatial micro-neighborhoods formally reconstruct continuous non-linear replicator vector fields without requiring sequential invasive biopsies. 2. **Identifiable Formulation on the Simplex:** We identify the exact algebraic collinearity between intrinsic fitness and interaction terms on the unit simplex, and formulate a fully identifiable 9-parameter absorbed payoff matrix ($\tilde{A}_{ij} = b_i + A_{ij}$) with condition number reduced by seven orders of magnitude (≈ 31–54 vs $\sim 4 \times 10^8$). 3. **Microenvironmental Decoupling & Spatial Controls:** We decouple hypoxia (Z_{hyp}) and vascular density (Z_{vasc}) confounders from cell proliferation, and benchmark spatial Delaunay interactions against linear regional composition (M_{2c}) and generic non-linear quadratics (M_{2d}), demonstrating a +7.26% gain in 5-fold spatial block cross-validation out-of-fold generalizability in breast carcinoma ($p < 1.11 \times 10^{-16}$) while establishing that lung SCC is predominantly governed by regional composition ($p = 0.266$). When evaluated directly on raw proliferation, joint regression accounts for 49.60%–54.30% of total tissue variance. 4. **Reproducibility Across Serial Slices:** Evaluated across independent consecutive serial sections ($\approx 10\,\mu$m apart), *InSituGame* demonstrates parameter concordance ($r = +0.811$, $p = 0.0079$), consistent game classification (Stag Hunt with tipping separatrix $x_{sep} \approx 0.80$), and blind out-of-sample predictive generalizability ($R^2_{\text{OOS}} = 10.20\%$ vs 6.21% for mass-action baseline; Diagnostic Odds Ratio = 2.30 [95% CI: 2.03–2.61], exact Fisher $p = 4.82 \times 10^{-12}$). 5. **Patient-Specific Game Individuality:** While the primary CytAssist breast IDC cohort exhibits Hawk-Dove coexistence ($x^* = 61.3\%$, 95% bootstrap CI: 0.526–0.725) and cutaneous melanoma displays suggestive Stag Hunt bistability ($x_{sep} = 48.1\%$, with bootstrap 95% CI spanning zero for $\tilde{\Delta}_R$), lung SCC is categorized as indeterminate / regional ($p = 0.266$). Fresh-frozen breast serial cuts independently display Stag Hunt bistability, demonstrating that evolutionary games are patient-specific ecological architectures that warrant individual biopsy profiling.

Open Science, Software, and Ethical Declarations

- **Software Implementation:** *InSituGame* is fully implemented as an open-source Python package (`insitugame`) with an automated test suite. All code, raw data streaming parsers, and figure generation scripts are publicly available on GitHub (https://github.com/Lukecele/spatial-game-assay) and permanently archived on Zenodo (https://doi.org/10.5281/zenodo.23585216) under the copyleft GNU General Public License v3.0 (GPL-3.0) to ensure unencumbered access for research institutions and hospitals.
- **Institutional Open Access Funding:** As an investigator affiliated with the School of Medicine, Università degli Studi di Napoli Federico II, open access publication is eligible for **0€ Article Processing Charges (APC)** through the national **CRUI-CARE (Conferenza dei Rettori delle Università Italiane)** Transformative Agreement with Oxford University Press.
- **Competing Interests:** The author declares no competing financial, personal, or commercial interests.
- **Originality:** This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere.

Suggested Independent Reviewers

To assist the editorial office in selecting qualified experts at the intersection of evolutionary game theory, mathematical oncology, and spatial omics, we suggest: 1. **Prof. Alexander R. A. Anderson, Ph.D.** — Chair, Integrated Mathematical Oncology (IMO), Moffitt Cancer Center, Tampa, FL, USA. Expert in evolutionary game theory, adaptive therapy, and spatial tumor modeling. Email: `Alexander.Anderson@moffitt.org` 2. **Prof. David Basanta, Ph.D.** — Associate Member, Integrated Mathematical Oncology (IMO), Moffitt Cancer Center, Tampa, FL, USA. Leading pioneer in evolutionary games in cancer ecosystems and stromal interactions. Email: `David.Basanta@moffitt.org` 3. **Artem Kaznatcheev, Ph.D.** — Mathematical Oncology & Evolutionary Game Theory, Oxford University / Moffitt Cancer Center. First author of the landmark 2019 *Nature Ecology & Evolution* paper on empirical cancer payoff matrices. Email: `artem.kaznatcheev@magd.ox.ac.uk` 4. **Prof. Carlo C. Maley, Ph.D.** — Arizona Cancer Evolution Center (ACE), Arizona State University, Tempe, AZ, USA. World leader in cancer evolutionary dynamics and clonal competition. Email: `maley@asu.edu`

Thank you for your consideration of our manuscript. We look forward to hearing from you.

Sincerely,

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